

Niharika Pant

(412) 689-3762 <https://www.linkedin.com/in/niharika-pant/> npant2@andrew.cmu.edu

EDUCATION

Carnegie Mellon University

Pittsburgh, PA, USA

Master of Science in Mechanical Engineering - Research

May 2027

Coursework: Robot Planning, Modern Control Theory, Intro to Human Robot Interaction, Advanced Mechatronic Design, Engineering Computation, Robot Dynamics and Analysis, Finite Element Analysis, AI/ML for Engineers.

Course Support: 24697 - Grader, Fall '25, 24694 - TA, Spring '26, 24780 - TA, Fall '26

City University of Hong Kong

Hong Kong

Bachelor of Engineering in Mechanical Engineering

May 2022

SKILLS

Robotics & ML: ROS/ROS2, MuJoCo, PyTorch, LoRA fine-tuning, imitation learning, control theory (state-space, PID)

Programming & CAD: C++, Python, MATLAB, Git, Fusion 360, OnShape, SolidWorks, KiCAD, FEA (Ansys)

Hardware & Manufacturing: 3D Printing, Soldering, Laser Cutting, Silicone Casting, Mill & Lathe Operation, Circuit Analysis

Systems: Technical Documentation, QC Protocols, Design Integration, Failure Analysis

EXPERIENCE

Robotic Caregiving and Human Interaction Lab - Carnegie Mellon University

Pittsburgh, PA, USA | January 2026 onwards

Graduate Research Student, advised by Prof. Zackory Erickson and Prof. Aviral Kumar

- Develop a hierarchical control system for assistive manipulation (high-level VLM for grasp strategy, low-level controller for arm motion) via $\pi 0.5$ fine-tuning, building on an ACT baseline.
- Assemble and configure teleoperated robotic systems, including wheelchair-mounted assistive manipulators, for data collection and policy evaluation; compare camera rig configurations to isolate effects on policy failure modes.
- Design and implement standardized data collection pipelines and experimental protocols for teleoperated manipulation datasets, including a physical mock-shelf testbed for grasp-strategy validation.
- Benchmark general-purpose VLMs (GPT, Claude, Gemini) on grasp-reasoning tasks over shelf imagery, identifying systematic grasp-direction biases; train and evaluate human-in-the-loop imitation learning policies, building on RaC and HG-Dagger.

Hanson Robotics

Hong Kong | September 2022 – July 2025

Humanoid Robotics Developer

- Engineered actuated humanoid facial systems with proprietary silicone skins for production of robots including Sophia, Grace and Mika, defining ROM limits, multi-servo coordination and viseme calibration for expressive facial motion.
- Reworked head production and calibration workflows, increasing throughput by ~60% and improving QC consistency.
- Diagnosed and resolved mechanical-electrical integration issues during deployments and demos; authored standardized assembly, calibration and testing documentation used to train engineers, technicians and interns.
- Provided technical operation and on-site support for presentations and showcases in Hong Kong and Shenzhen.

Hanson Robotics

Hong Kong | December 2020 – April 2022

Mechanical Design Research Fellow

- Designed and validated robotic arm components in SolidWorks using structural and material testing.
- Optimized humanoid skull assemblies and fabricated silicone skin components through proprietary manufacturing processes.
- Prototyped a chest-mounted camera module to expand robot visual coverage.

PROJECTS

CMU Lunabotics, Team Moon Miners - Mechanical Engineering Team Member (Mobility sub-team)

- Designed and prototyped a track-based mobility system for a lunar excavation rover, integrating sheet metal components, COTS hardware, 3D-printed tracks, sprockets and motor housings.
- Developed custom test rigs to characterize traction, slip, drawbar pull and payload capacity; analyzed results alongside competitor rover architectures to inform mobility design recommendations.
- Contributed to a first-year team's autonomy-scoring milestone and Best Presentation and Demonstration award.

Aarhus University - Overseas Research Intern

- Studied harmonic drive mechanisms for robotic arm actuation under Prof. Xuping Zhang.
- Co-developed a ROS-based digital twin for object manipulation and system bottleneck analysis.

CityU Rocket Project Team - Design Engineer

- Designed EDF rocket subsystems (housing, fins, landing components) in AutoCAD; validated 3D-printed parts via stress, drop and strain testing to optimize material and print parameters.

PUBLICATIONS

S. Kowshik, S. Venkataraman, L. Wang, **N. Pant**, M. Simchowitz, and A. Kumar, "Adaptation of Generalist Robot Policies with Minimal Data," *RL4VLA Workshop, Robotics: Science and Systems (RSS)*, 2026. (Accepted)